Hydraulic Enceladus

COSC-P project

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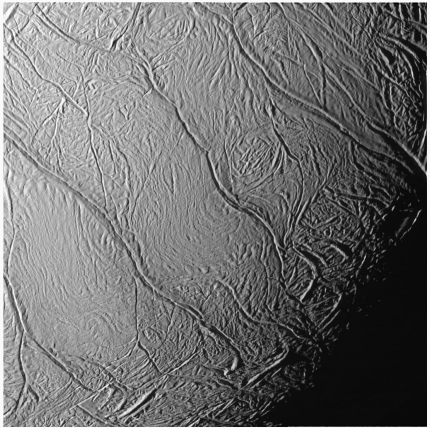
Motivation

Icy Enceladus

- **Enceladus**, one of Saturn's moons, hides a subsurface ocean beneath an ice shell.
- so called *Tiger stripes*, *i.e.* cracks in the ice shell at the south pole *eject* plumes of water vapor and ice grains into space - **how?**
- Saturn exerts **tidal forces** on Enceladus $\Longrightarrow$ ice shell squeezes and stretches $\Longrightarrow$ the water in the cracks gets displaced
- since the orbit is periodic, the motion of the ice shell is cyclic as well



Tiger stripes. Source: NASA/JPL

Problem settings

Modelling choices

- the crevasses are deep narrow *effectively* rectangular domains,
- the fluid is water, *i.e.* **incompressible** Newtonian[1], $\rho = 1000\,\mathrm{kg/m^3}$, $\mu = 1.5\,\mathrm{mPa\,s}$,
- the only body force is **gravity** $g = 0.113\,\mathrm{m/s^2}$,
- the fluid can *freely slip* along the walls,
- the motion of the walls is *harmonic*

[1] Much higher viscosity has been used in the simulation.

Governing equations

We are solving for $(\mathbf{v}, p, \mathbf{u})$ with $\Omega_0 = (0,1) \times (0,10)$ given such that $\forall t \in (0,T), T > 0$ inside the domain

Governing equations

$$\Omega_t = \left\{ \mathbf{x} + \mathbf{u}(t, \mathbf{x}) \mid \mathbf{x} \in \Omega_0 \right\} \subset \mathbb{R}^2$$

it holds $\forall t \in (0,T)$

$$\begin{aligned} \Delta \mathbf{u} &= 0, \\ \rho \left(\partial_t \mathbf{v} + (\mathbf{v} - \partial_t \mathbf{u}) \cdot \nabla \mathbf{v} \right) &= \nabla \cdot \mathbb{T} + \rho \mathbf{g}, \\ \nabla \cdot \mathbf{v} &= 0, \\ \mathbb{T} &= -p\mathbb{I} + 2\mu \mathbb{D}. \end{aligned}$$

Boundary conditions

Denote $\partial\Omega_t = \Gamma_{\text{bot}} \cup \Gamma_{\text{left}} \cup \Gamma_{\text{right}} \cup \Gamma_{\text{top}}$. The trio $(\mathbf{v}, p, \mathbf{u})$ must satisfy on $(0, T) \times \partial\Omega_t$

Boundary conditions

$$
\begin{aligned}
u_x &= u_0 \sin(2\pi f t), \partial_x u_y = 0, \text{ on } \Gamma_{\text{left}}, \\
u_x &= -u_0 \sin(2\pi f t), \partial_x u_y = 0 \text{ on } \Gamma_{\text{right}}, \\
u_y &= 0, \partial_y u_x = 0 \text{ on } \Gamma_{\text{bot}}, \\
v_x &= 2u_0 \pi f \cos(2\pi f t), \partial_x v_y = 0, \text{ on } \Gamma_{\text{left}}, \\
v_x &= -2u_0 \pi f \cos(2\pi f t), \partial_x v_y = 0, \text{ on } \Gamma_{\text{right}}, \\
v_y &= 0, \partial_y v_x = 0 \text{ on } \Gamma_{\text{bot}}, \\
\partial_t \mathbf{u} &= \mathbf{v}, \text{ on } \Gamma_{\text{top}}, \\
\mathbb{T}\mathbf{n} &= \mathbf{0}, \text{ on } \Gamma_{\text{top}}.
\end{aligned}
$$

Numerical solution

Implementation strategy

To solve the problem numerically, we use the *FEniCS* library with the choices:

- **Discretization:** (P_2, P_1, P_2) elements for $(\mathbf{v}, p, \mathbf{u})$ *i.e.* Hood - Taylor for fluid, P_2 for mesh,
- **Coupling**: full ALE method with the monolithic formulation for $(\mathbf{v}, p, \mathbf{u})$.
- **Kinematic top boundary condition**: (symmetric) Nitsche Method for weak enforcement of the free surface BC
- **Time Integration:** implicit Crank-Nicolson θ-scheme
- **Solver:** Newton-based with MUMPS direct solver.

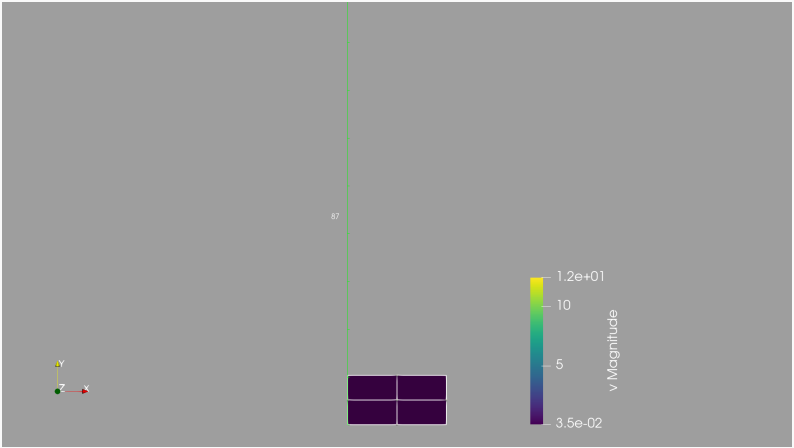
Parameters

The following parameters have been chosen to demonstrate the **qualitative behaviour** of the model

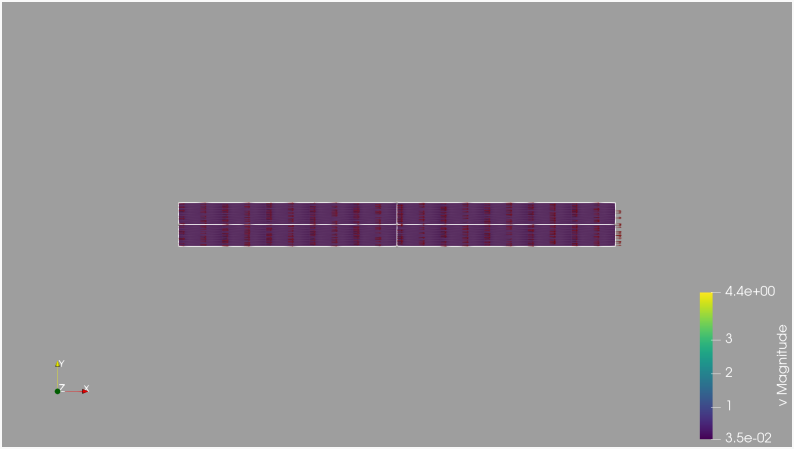
- *time step*: $dt = 0.05$
- *mesh resolution*: 10 cells in horizontal direction and 400 cells in vertical direction
- *frequency of wall motion*: $f = 0.05\,\mathrm{s}^{-1}$,
- *amplitude of wall motion*: $u_0 = 0.40\,\mathrm{m}$,
- *gravitational acceleration*: $g = 0.113\,\mathrm{m/s}^2$,
- *density*: $\rho = 1000\,\mathrm{kg/m}^3$,
- *dynamic viscosity*: $\mu = 1490\,\mathrm{Pa\,s}$,

Results

Complete view



Close view



Summary

- Within our approach we were able to obtain the maximum squeeze $u_0 = 0.4$, *i.e.* narrowing the region down to **20 %** of its original width
- The simulation validated the hypothesis: squeezing can induce **significant vertical displacement.**
- To achieve that, ***significant* increase** in viscosity ought to be introduced for stability: $\mu = 1490$ Pa s versus $\mu = 1.49$ mPa s
- Note that the *timescales* are **not realistic**. Solution to the full problem would require *much longer simulation times*; the periods of the motions are in fact in the magnitude of **hours**

Source: xkcd.com/2267